



Colorado Mesh Monthly

August & September 2026

Your once-a-month digest of everything on the mesh — so you don't have to read every message.

Hey folks, and welcome to the very first Colorado Mesh Monthly. Since this is our kickoff and we're publishing mid-month, we're reaching back across both August and September so nobody misses the good stuff. Going forward this lands once a month — but for now, settle in: it was a huge two months. We linked the mesh across state lines into Nebraska, kept pushing the Denver metro onto MediumFast, started seriously mapping out regional "scopes," and welcomed a genuinely wonderful flood of new operators. Here's the recap, sorted by protocol so you can jump straight to your corner of the mesh.

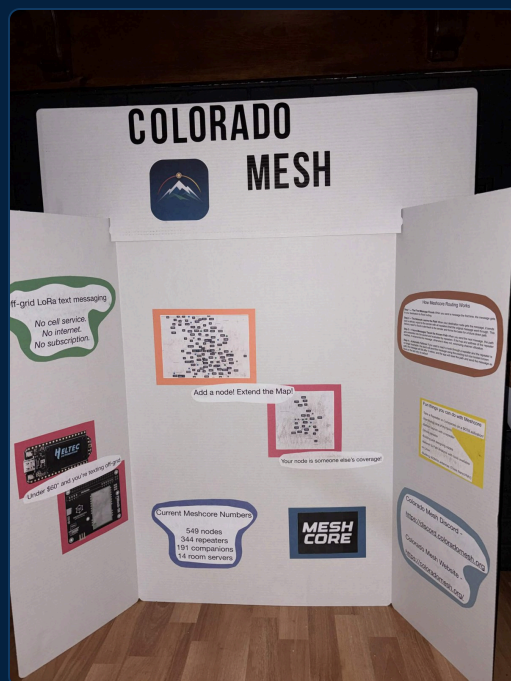
General News

- **We crossed over 1,000 Discord members**, and the pace of new arrivals hasn't slowed. Between August and September the introductions channel was one of the busiest in the whole server. If you're new: welcome, and don't be shy about asking where to put your first node.
- **A community reminder on airtime etiquette.** As the network grows, the public channels saw a spike in automated traffic — bots broadcasting every few seconds, plus the occasional commercial or crypto spam. Rin (R-4) asked everyone to keep automated messages to no more than once every few hours (a once-a-morning weather post is fine; every 30 minutes is not), and to keep commercial broadcasts off the public channels entirely. LoRa airtime is tiny and shared by everyone, so a little restraint keeps the whole mesh usable.
- **A community configuration tool consolidation.** The group standardized on <https://tools.meshcore.coloradomesh.org> as the primary config tool and deprecated the older site, streamlining how people set up and manage nodes.
- **A town hall may be in the works.** Zeva floated the idea of an online town hall to make up for the lack of recent in-person gatherings, and JohnnyW5KV has been vocal about wanting regular meetups. Nothing's scheduled yet — but if the interest is there, say so in the channels and help shape the format.
- **Coverage keeps pushing west.** More repeaters are going up around Grand Junction and the western slope, steadily filling in a part of the state that was thin not long ago. If you're out that way, an observer or a well-placed node goes a long way — coordinate in `#western-slope`.
- **New areas, more voices.** Traffic has been strong across the regional channels, and the community keeps carving out new regional homes as the map fills in. Whichever corner of Colorado you're in, there's probably a channel for it now.

- **BARCFest — October 4 (Boulder Amateur Radio Club).** Colorado Mesh has **two tables reserved** at BARCFest, the Boulder Amateur Radio Club's hamfest — come find us, meet the people behind the callsigns, and see the gear in person. There's also been talk of asking BARC about hosting a mesh repeater on one of their tower sites, so it may be more than a social visit. Details: <https://barcw0dk.wordpress.com>
- **We had a strong showing at the RMHAM Summer Swapfest (Aug 23)** — a good turnout at our tables and a lot of great conversations with folks new to the mesh. Thanks to everyone who staffed it.



Colorado Mesh at the RMHAM Summer Swapfest, Aug 23 (photo: JohnC)



The Colorado Mesh table at the Swapfest (photo: JohnnyW5KV)

- **Parker Radio Association** held its monthly meeting on September 7 with a presentation on AllStar (via JohnnyW5KV). Keep an eye on announcements for the next one.
- **Weekly Net — every Thursday.** There's no dedicated channel: just post to the MeshCore Public channel any time on Thursday and you're checked in. New this cycle, Old Man Malice wired the net into MeshWars so your check-in earns credit there too.



On the summit of Mt Blue Sky (14,130 ft), M0TH3R logged which repeaters heard — and were heard by — a node at the top of Colorado. Photo: M0TH3R, Aug 31. [Two landscape options (B/C) and a portrait (A) are in issues/2026-09/photos/contact-sheet.html — open it in a browser to switch.]

Meshtastic

- **The MediumFast migration is the big Meshtastic story.** The Denver metro is steadily moving from LongFast to MediumFast for better performance, and Lookout Mountain shifted its presets to nudge people toward MediumFast. Adoption is still uneven — plenty of nodes outside the metro remain on LongFast, and there's healthy discussion about the range tradeoffs — so check what your area is actually running before you switch. Legacy LongFast bridging is being wound down. (Runr, TickleMeTemplar, nwithan8) <https://coloradomesh.org/news/mediumfast-changeover>
- **A cross-protocol bridge experiment.** nwithan8 ran a live test bridging Meshtastic (LongFast) and MeshCore (Public) in the Denver area to see what actually crosses between the networks. The finding: Meshtastic traffic in range was light enough that there wasn't much worth bridging — useful data for how (and whether) to link the two going forward.
- **Firmware fix worth updating for.** A Meshtastic firmware update this cycle addressed a man-in-the-middle issue — a good reason to make sure your nodes are current. Discussion in #meshtastic.
- **Know your presets.** Alongside MediumFast, there was ongoing talk about LongTurbo and frequency-slot choices, and what each means for range and congestion. If you're unsure what to run, ask before you flash — the channel is happy to help.
- **MQTT, maps, and observers.** A lot of quiet-but-useful plumbing work continued: publishing LongFast/MediumFast traffic to MQTT topics, feeding the Meshtastic map and analyzer, and standing up observers. If you want your area on the map, running an observer is the way in.
- **Work on the Lookout bridge.** omgitsgela has been configuring a bridge up on Lookout Mountain — a WiFi repeater for remote access, MQTT bridging, and local mirroring of MediumFast and LongFast, with a second radio planned to add LongFast support. It's a meaningful step toward tying the high-site infrastructure together, with the door open to adding Reticulum up there down the road.

MeshCore

- **We're meshing across state lines.** A working path along I-76 now links the Colorado mesh to Nebraska — an advert from a node near DIA was heard as far as Ogallala. It's a genuine milestone: coverage has grown from a metro network into something that reaches past the state border. (nwithan8)
- **"Scopes" — defining our regions.** With the network sprawling, a new #scopes channel spun up to plan and standardize MeshCore region definitions. To stay compatible with our existing MeshMapper regioning, the approach layers **geographic, county, and airport IATA codes together** rather than swapping one for another — abandoning IATA would mean redoing everything from the ground up. nwithan8 has been coordinating with Nebraska, Utah, and Wyoming so the scheme works across state lines. It's an active, much-debated topic — jump in with input before anything is locked in. (Rin R-4, Zeva, nwithan8)
- **Observers are filling in the map.** New observers came online across the state — including Grand Junction — feeding the MQTT relay and the analyzer so we can actually see how the network is growing. A naming tool was added to help operators label nodes to convention, and folks in under-covered regions are encouraged to stand up an observer and report in. (nwithan8, Zeva)
- **Testing toward a 500 kHz profile.** There's active experimentation to settle on a good 500 kHz-wide MeshCore profile (folks have been comparing notes around 915 MHz, with Philly Mesh testing near 915.250). Nothing's finalized yet — it's one of the most-debated topics on the mesh right now, so if you want a say in the settings, follow along in #meshcore.
- **The MQTT relay ties it together.** A service on a VPS reads the MQTT broker and relays MeshCore traffic into Discord, so observers' packets show up where the community can see them — part of what makes the analyzer and maps useful.
- **Mind the naming convention.** As nodes multiply, sticking to the MeshCore naming convention keeps the map legible for everyone — worth a look if you're renaming or standing up a repeater.
- **New repeaters keep lighting up.** Among them, mrpatzy placed one near Brittany Hill with strong line-of-sight to Denver and the mountains, and more are planned around Grand Junction. Every well-placed node makes the whole mesh more reliable.

Reticulum

- **Our RRC hub is drawing people from all over.** The Colorado Mesh **RRC (Reticulum Relay Chat)** server has become a genuinely popular gathering spot — folks are joining from around the world to hang out, and a lot of the Reticulum conversation actually happens *there* rather than in Discord. A big shout-out to **KK4FRN (Alex)** for spinning up the hub. Join us: #reticulum.
- **You don't need RF gear to try Reticulum.** This is the easiest on-ramp in the whole hobby: with just an internet connection you can join the RRC hub and get a feel for Reticulum before buying a single radio. **M3SHGHØST** added a Reticulum quick-setup guide to the Mesh Client to make that first step painless. Give it a spin and come say hi.
- **The Mesh Client is a great Reticulum window.** Beyond RRC chat, it now gives you NomadNet browsing — and, new this cycle, *editing* — all in one app. **Runr** and others have been experimenting with a range of Reticulum mobile apps on Android and iOS too, so there are more ways than ever to get on Reticulum from your pocket.



Gear & Firmware

New devices, firmware, and the occasional "don't do that" bulletin from the workbench channels this cycle:

- **Safety bulletin — never remove the antenna on a powered board.** A good reminder from Packman5280: if a LoRa board powers up (or stays powered) with no antenna attached, you can burn out the RF power amp. Power down *before* you swap antennas. Cheap mistake, expensive board.
- **Firmware fix for RNode over BLE.** An update this cycle fixed Bluetooth for RNode on ESP/nRF boards (V3, V4, T-Echo, T114). If you run RNode and BLE's been flaky, grab the update. (dude.eth, #software)
- **Meshtastic firmware fixed a man-in-the-middle issue** — worth updating your nodes for. (See the Meshtastic section above.)
- **Custom repeater firmware from nightcrawler.** A standalone build that bakes in the advert/policy filtering from OpenHop — shared for anyone who wants to try it. (#meshcore)
- **Board note — the RAK 1W is a lot of board for the money.** JohnnyW5KV gives it a +1: integrated input/output filters, solid little board. A popular pick for repeaters right now.
- **OTA gotcha.** A couple of folks learned the hard way that an over-the-air update on a rooftop/solar node can leave you climbing up to re-flash if it doesn't take — worth having a wired fallback plan before you push firmware to anything hard to reach.



Get the Mesh Client

New here, or still juggling separate apps? The community maintains **Mesh Client** — a free, cross-platform desktop app (Windows, macOS, Linux) that speaks Meshtastic, MeshCore, *and* Reticulum from one window, so you don't need a different tool for each protocol. It's actively developed — a big thanks to Joey (NV0N), its lead developer, and the contributors — and it's the easiest way to manage your nodes and follow the mesh from a real screen. Grab it at github.com/Colorado-Mesh/mesh-client and give it a spin.



For Sale & Wanted

- No member classifieds came in this cycle. A couple of build resources worth a look, though: Zeva shared a Printables model for a 1-watt low-profile solar mesh car node, and pointed to the RAK3401 1W LoRa booster kit on the RAKwireless store. _Got gear to sell or a part you're hunting for? Drop it in the channel and we'll list it here next month._

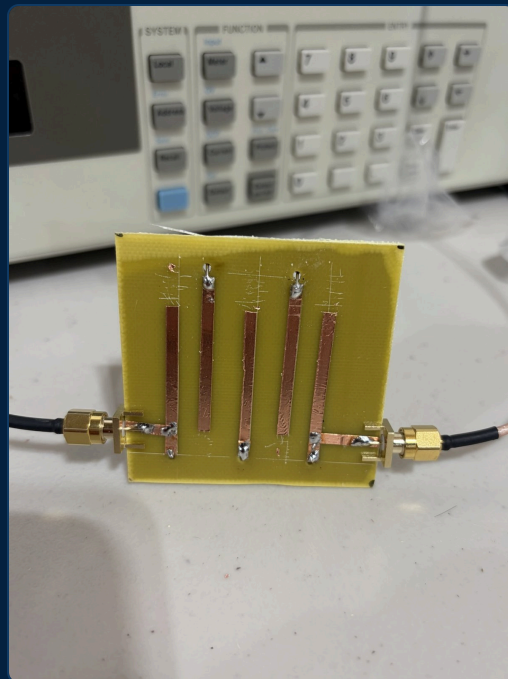
Call to Action

We need Reticulum full repeaters in high spots — build yours now! High-elevation nodes are what turn scattered coverage into a real network, and Reticulum full transport nodes on ridges and rooftops are exactly where we're thin. If you've got a good vantage point, tower access, or a solar setup you've been meaning to deploy, this is the single highest-leverage thing you can do for the mesh right now. Ask in the channels — budt WORMT and others have offered to help with repeater setup, and the wiki has the configuration docs.

Run an observer in your area — especially outside the metro. Observers are how a region gets onto the map and the analyzer. If you're in a corner of the state that looks empty on the map, that's often just because nobody's reporting from there yet. The wiki has a guide to running an observer, and it's a low-effort, high-value way to contribute.

Community Spotlight

Shout-out to **Iguy**, who stood up a brand-new repeater near Lexington and Union in Colorado Springs — balcony-mounted, solar plus battery, at 6,810 ft. That "just put one up and see what happens" energy is exactly how the mesh grows, one rooftop at a time. And a nod to **Zeva**, whose fingerprints are all over this cycle: region definitions, observer onboarding, config tooling, and a steady stream of technical help in the channels. Welcome to the map, Iguy — and thanks, Zeva.



beala's hand-built 900 MHz bandpass filter (#projects-and-builds)

Bonus build spirit of the month: beala hand-fabricated a 900 MHz bandpass filter from a PCB and copper tape — the VNA showed a real filter, just with (beala's words) a hilarious 33 dB of insertion loss. Not shippable yet, but exactly the kind of tinkering that makes this community fun.

More folks who moved the mesh forward this cycle: **JohnnyW5KV**, who finished a solar install and a chimney-mast setup for a Station G3 and has been a fixture troubleshooting repeaters; **EnderW**, who's been building out the northwest corner of the state; and **nightcrawler**, who shared RF sweep results and custom repeater firmware. The mesh grows on exactly this kind of hands-on effort — thank you all.



Around the Web

beala's writeup on the FCC and using 500 kHz bandwidth — a genuinely good technical deep-dive: <https://beala.substack.com/p/the-fcc-want-me-to-use-more-bandwidth>

JohnnyW5KV's video reviews of mesh gear (the PeakMesh Climber, among others) on YouTube.

A new coloradomesh.org news page on the 2-byte path changeover is in progress (KFØUFO - Andrew) — <https://coloradomesh.org/news/2-byte>

Colorado MeshCore analyzer & tools — network stats, beacon data, contact-pack downloads, and range checks: <https://analyzer.meshcore.coloradomesh.org> and <https://tools.meshcore.coloradomesh.org>

Setting up Reticulum infra? Zeva shared a Docker config for a NomadNet transport node in #reticulum — a handy starting point.



Story of the Month: Meshing the Divide

If these two months had a single theme, it was *reach*. Back in August, the network was still largely a Front Range affair — a cluster of nodes trading packets between the foothills and the metro. By mid-September, the map told a different story: messages were routinely finding their way to the far corners of the state and, for the first time, spilling clean across a state line (you'll find the specifics up in the MeshCore section).

What's worth sitting with isn't any one hop — it's how it happened. Nobody flipped a switch. It was the cumulative payoff of dozens of small, unglamorous acts: a node zip-tied to a balcony, an observer quietly reporting from a town nobody had covered yet, a preset nudged on a mountaintop, and an awful lot of people comparing notes about antennas and coax loss at 900 MHz. That's the whole trick to a community mesh — it doesn't scale by decree, it scales by people deciding, one rooftop at a time, that their corner of Colorado belongs on the map. Not bad for a pile of low-power radios and a shared stubbornness about coverage.



Until Next Month

That's the catch-up. From here we'll land once a month, covering the prior month. If you lit up a repeater, printed an enclosure, ran a great wardrive, or finally beat a stubborn SNR — tell us, so it lands in the next issue. Big or small, we want it.

Nerd Joke of the Month: Why did the LoRa packet bring a ladder to the net? It heard the best check-ins come from *high* SNR.

The Prize: this month's giveaway is a **RAKwireless WisBlock Meshtastic Starter Kit (US\$915)** — the RAK19007 Base + RAK4631 Core — bundled with a **RAK12501 GNSS location tracker module**. Everything you need to build a GPS-equipped Meshtastic node from scratch. Starting next month we'll draw a random winner from everyone who reacts to the newsletter in Discord — so react to this issue to be entered!



Sources & Further Reading

1. Colorado MeshCore analyzer / live map — <https://analyzer.meshcore.coloradomesh.org>
2. Meshtastic live map — <https://map.meshtastic.coloradomesh.org>
3. MeshCore config tools — <https://tools.meshcore.coloradomesh.org>
4. MediumFast changeover — <https://coloradomesh.org/news/mediumfast-changeover>
5. 2-byte path changeover — <https://coloradomesh.org/news/2-byte>
6. Colorado MeshCore Regions (wiki) — https://wiki.coloradomesh.org/wiki/Colorado_MeshCore_Regions
7. MeshCore MQTT / running an observer (wiki) — https://wiki.coloradomesh.org/wiki/MeshCore_MQTT
8. Weekly Net check-in — <https://weekly-net.meshcore.coloradomesh.org>
9. Donate to keep the network running — <https://coloradomesh.org/donate>
10. Join / getting started — <https://coloradomesh.org>